

AI-POWERED CKD DIAGNOSIS: ENHANCING HOSPITAL DECISION SUPPORT SYSTEMS

Total number of rows - **400**, columns - **5**

To calculate the percentage of correct classifications (i.e., accuracy) for predicting buyers or non-buyers, you can use the following formula:

PARAMS	BUYER	NON-BUYER	MACRO AVG	WEIGHTED AVG
Precision	$(\text{True Buyer} / (\text{True Buyer} + \text{False Non-Buyer}))$	$(\text{True Non-Buyer} / (\text{True Non-Buyer} + \text{False Buyer}))$	$(\text{Buyer Precision} + \text{Non-Buyer Precision}) / 2$	$((\text{Precision Buyer} * \text{Buyer Users}) / \text{No of Users}) + ((\text{Precision Non-Buyer} * \text{Non-Buyer Users}) / \text{No of Users})$
Recall	$(\text{True Buyer} / (\text{True Buyer} + \text{False Buyer}))$	$(\text{True Non-Buyer} / (\text{True Non-Buyer} + \text{False Non-Buyer}))$	$(\text{Buyer Recall} + \text{Non-Buyer Recall}) / 2$	$((\text{Recall Buyer} * \text{Buyer Users}) / \text{No of Users}) + ((\text{Recall Non-Buyer} * \text{Non-Buyer Users}) / \text{No of Users})$
F1 Score	$2 * (\text{Precision Buyer} * \text{Recall Buyer}) / (\text{Precision Buyer} + \text{Recall Buyer})$	$2 * (\text{Precision Non-Buyer} * \text{Recall Non-Buyer}) / (\text{Precision Non-Buyer} + \text{Recall Non-Buyer})$	$(\text{F1-Score Buyer} + \text{F1-Score Non-Buyer}) / 2$	$((\text{F1-Score Buyer} * \text{Buyer Users}) / \text{No of Users}) + ((\text{F1-Score Non-Buyer} * \text{Non-Buyer Users}) / \text{No of Users})$
Accuracy	$((\text{True Buyer} + \text{True Non-Buyer}) / (\text{True Buyer} + \text{False Buyer} + \text{True Non-Buyer} + \text{False Non-Buyer}))$			

SUPPORT VECTOR MACHINE CLASSIFICATION

Confusion Matrix: [[82 3] [26 23]]

Buyer	True is 82	False is 3
Non-Buyer	False is 26	True is 23

The SVM algorithm can be used to calculate the classification accuracy for predicting buyers or non-buyers.

Params	Buyer	%	Non-buyer	%	Macro AVG	%	Weighted AVG	%
Precision	$(82/(82+26))$	0.76	$(23/(23+3))$	0.88	$(0.76+0.88)/2$	0.82	$((0.76*85)/134)+((0.88*49)/134)$	0.8
Recall	$(82/(82+3))$	0.96	$(23/(23+26))$	0.47	$(0.96+0.47)/2$	0.72	$((0.96*85)/134)+((0.47*49)/134)$	0.78
F1 Score	$2*(0.76*0.96)/(0.76+0.96)$	0.85	$2*(0.88*0.47)/(0.88+0.47)$	0.61	$(0.85+0.61)/2$	0.73	$((0.85*85)/134)+((0.61*49)/134)$	0.76
Accuracy	$((82+23)/(82+3+26+23))$							0.78

DECISION TREE CLASSIFICATION

Confusion Matrix: [[76 9] [8 41]]

Buyer	True is 76	False is 9
Non-Buyer	False is 8	True is 41

The DT algorithm can be used to calculate the classification accuracy for predicting buyers or non-buyers.

Params	Buyer	%	Non-buyer	%	Macro AVG	%	Weighted AVG	%
Precision	$(76/(76+8))$	0.9	$(41/(41+9))$	0.82	$(0.9+0.82)/2$	0.86	$((0.9*85)/134)+((0.82*49)/134)$	0.87
Recall	$(76/(76+9))$	0.89	$(41/(41+8))$	0.84	$(0.89+0.84)/2$	0.87	$((0.89*85)/134)+((0.84*49)/134)$	0.87
F1 Score	$2*(0.9*0.89)/(0.9+0.89)$	0.89	$2*(0.82*0.84)/(0.82+0.84)$	0.83	$(0.89+0.83)/2$	0.86	$((0.89*85)/134)+((0.83*49)/134)$	0.87
Accuracy	$((76+41)/(76+9+8+41))$							0.87

RANDOM FOREST CLASSIFICATIONConfusion Matrix: $\begin{bmatrix} 78 & 7 \\ 6 & 43 \end{bmatrix}$

Buyer	True is 78	False is 7
Non-Buyer	False is 6	True is 43

The RF algorithm can be used to calculate the classification accuracy for predicting buyers or non-buyers.

Params	Buyer	%	Non-buyer	%	Macro AVG	%	Weighted AVG	%
Precision	$(78/(78+6))$	0.93	$(43/(43+7))$	0.86	$(0.93+0.86)/2$	0.9	$((0.93*85)/134) + ((0.86*49)/134)$	0.9
Recall	$(78/(78+7))$	0.92	$(43/(43+6))$	0.88	$(0.92+0.88)/2$	0.9	$((0.92*85)/134) + ((0.88*49)/134)$	0.91
F1 Score	$2*(0.93*0.92)/(0.93+0.92)$	0.92	$2*(0.86*0.88)/(0.86+0.88)$	0.87	$(0.92+0.87)/2$	0.9	$((0.92*85)/134) + ((0.87*49)/134)$	0.9
Accuracy	$((78+43)/(78+7+6+43))$							0.9

LOGISTIC CLASSIFICATIONConfusion Matrix: $\begin{bmatrix} 85 & 0 \\ 49 & 0 \end{bmatrix}$

Buyer	True is 85	False is 0
Non-Buyer	False is 49	True is 0

The Logistic algorithm can be used to calculate the classification accuracy for predicting buyers or non-buyers.

Params	Buyer	%	Non-buyer	%	Macro AVG	%	Weighted AVG	%
Precision	$(85/(85+49))$	0.63	$(0/(0+0))$	0	$(0.63+0)/2$	0.32	$((0.63*85)/134) + ((0*49)/134)$	0.4
Recall	$(85/(85+0))$	1	$(0/(0+49))$	0	$(1+0)/2$	0.5	$((1*85)/134) + ((0*49)/134)$	0.63
F1 Score	$2*(0.63*1)/(0.63+1)$	0.77	$2*(0*0)/(0+0)$	0	$(0.77+0)/2$	0.39	$((0.77*85)/134) + ((0*49)/134)$	0.49
Accuracy	$((85+0)/(85+0+49+0))$							0.63

KNN CLASSIFICATION

Confusion Matrix: [[74 11] [14 35]]

Buyer	True is 74	False is 11
Non-Buyer	False is 14	True is 35

The KNN algorithm can be used to calculate the classification accuracy for predicting buyers or non-buyers.

Params	Buyer	%	Non-buyer	%	Macro AVG	%	Weighted AVG	%
Precision	$(74/(74+14))$	0.84	$(35/(35+11))$	0.76	$(0.84+0.76)/2$	0.8	$((0.84*85)/134) + ((0.76*49)/134)$	0.81
Recall	$(74/(74+11))$	0.87	$(35/(35+14))$	0.71	$(0.87+0.71)/2$	0.79	$((0.87*85)/134) + ((0.71*49)/134)$	0.81
F1 Score	$2*(0.84*0.87)/(0.84+0.87)$	0.85	$2*(0.76*0.71)/(0.76+0.71)$	0.61	$(0.85+0.61)/2$	0.73	$((0.85*85)/134) + ((0.61*49)/134)$	0.76
Accuracy	$((74+35)/(74+11+14+35))$							0.81

NAIVE BAYE'S CLASSIFICATION**Naive Baye's MultinomialNB**

Confusion Matrix: [[85 0] [49 0]]

Buyer	True is 85	False is 0
Non-Buyer	False is 49	True is 0

The NB MultinomialNB algorithm can be used to calculate the classification accuracy for predicting buyers or non-buyers.

Params	Buyer	%	Non-buyer	%	Macro AVG	%	Weighted AVG	%
Precision	$(85/(85+49))$	0.63	$(0/(0+0))$	0	$(0.63+0)/2$	0.32	$((0.63*85)/134) + ((0*49)/134)$	0.4
Recall	$(85/(85+0))$	1	$(0/(0+49))$	0	$(1+0)/2$	0.5	$((1*85)/134) + ((0*49)/134)$	0.63
F1 Score	$2*(0.63*1)/(0.63+1)$	0.77	$2*(0*0)/(0+0)$	0	$(0.77+0)/2$	0.39	$((0.77*85)/134) + ((0*49)/134)$	0.49
Accuracy	$((85+0)/(85+0+49+0))$							0.63

Naive Baye's BernoulliNB**Confusion Matrix:** $\begin{bmatrix} 85 & 0 \\ 49 & 0 \end{bmatrix}$

Buyer	True is 85	False is 0
Non-Buyer	False is 49	True is 0

The NB BernoulliNB algorithm can be used to calculate the classification accuracy for predicting buyers or non-buyers.

Params	Buyer	%	Non-buyer	%	Macro AVG	%	Weighted AVG	%
Precision	$(85/(85+49))$	0.63	$(0/(0+0))$	0	$(0.63+0)/2$	0.32	$((0.63*85)/134) + ((0*49)/134)$	0.4
Recall	$(85/(85+0))$	1	$(0/(0+49))$	0	$(1+0)/2$	0.5	$((1*85)/134) + ((0*49)/134)$	0.63
F1 Score	$2*(0.63*1)/(0.63+1)$	0.77	$2*(0*0)/(0+0)$	0	$(0.77+0)/2$	0.39	$((0.77*85)/134) + ((0*49)/134)$	0.49
Accuracy	$((85+0)/(85+0+49+0))$							0.63

Naive Baye's CategoricalNB**Confusion Matrix:** $\begin{bmatrix} 82 & 3 \\ 18 & 31 \end{bmatrix}$

Buyer	True is 82	False is 3
Non-Buyer	False is 18	True is 31

The NB CategoricalNB algorithm can be used to calculate the classification accuracy for predicting buyers or non-buyers.

Params	Buyer	%	Non-buyer	%	Macro AVG	%	Weighted AVG	%
Precision	$(82/(82+18))$	0.82	$(31/(31+3))$	0.91	$(0.82+0.91)/2$	0.87	$((0.82*85)/134) + ((0.91*49)/134)$	0.85
Recall	$(82/(82+3))$	0.96	$(31/(31+18))$	0.63	$(0.96+0.63)/2$	0.8	$((0.96*85)/134) + ((0.63*49)/134)$	0.84
F1 Score	$2*(0.82*0.96)/(0.82+0.96)$	0.88	$2*(0.91*0.63)/(0.91+0.63)$	0.74	$(0.88+0.74)/2$	0.81	$((0.88*85)/134) + ((0.74*49)/134)$	0.83
Accuracy	$((82+31)/(82+3+18+31))$							0.84

Naive Baye's ComplementNB**Confusion Matrix:** [[40 45] [20 29]]

Buyer	True is 40	False is 45
Non-Buyer	False is 20	True is 29

The NB ComplementNB algorithm can be used to calculate the classification accuracy for predicting buyers or non-buyers.

Params	Buyer	%	Non-buyer	%	Macro AVG	%	Weighted AVG	%
Precision	$(40/(40+20))$	0.67	$(29/(29+45))$	0.39	$(0.67+0.39)/2$	0.53	$((0.67*85)/134) + ((0.39*49)/134)$	0.57
Recall	$(40/(40+45))$	0.47	$(29/(29+20))$	0.59	$(0.47+0.59)/2$	0.53	$((0.47*85)/134) + ((0.59*49)/134)$	0.51
F1 Score	$2*(0.67*0.47)/(0.67+0.47)$	0.55	$2*(0.39*0.59)/(0.39+0.59)$	0.47	$(0.55+0.47)/2$	0.51	$((0.55*85)/134) + ((0.47*49)/134)$	0.52
Accuracy	$((40+29)/(40+45+20+29))$							0.51

Result:

The **Random Forest Classification** algorithm gives the best result with an accuracy of **0.9**.